

Script-Controlled ACL – Restrict Record Access Based on Field Value

Naan Mudhalvan / Skill Wallet Project - Requirement Analysis Phase Document

1. Project & Team Information

- Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value
- Phase: Requirement Analysis Phase
- Team ID: SWTID-2026-9611
- Team Leader: Adhsa Rani P
- Team Members: Gayathri A, Parkavi P, Priyadharshini M, Chinthanika P
- Platform: ServiceNow / SkillWallet (SmartBridge)

2. Business Requirement & Objectives

- Business Need: Organization requires granular data governance where access to institutional records is controlled dynamically based on field values (e.g., branch, state, security classification) rather than static role assignments alone.
- Core Objectives:
 1. Enforce row-level security dynamically on the `u_institution_details` table.
 2. Implement script-controlled evaluation for READ, WRITE, CREATE, and DELETE operations.
 3. Restrict DELETE access specifically based on field value criteria (e.g., `u_branch == 'EEE'`).
 4. Ensure zero-trust security without breaking application usability or list view performance.

3. Functional Requirements

- FR-01 (Role Mapping): Map custom security roles (bb1, bb2, bb3, bb4) to user categories.
- FR-02 (Field-Based Access): Evaluate record field values (`u_branch`, `state`, `is_confidential`) during runtime access requests.
- FR-03 (CRUD Governance):
 - READ: Users can view records relevant to their assigned branch/department.
 - CREATE: Users with role `bb2` or higher can create new records.
 - WRITE: Users with role `bb3` or higher can update non-locked records.
 - DELETE: Only users with role `bb4` can delete records, strictly limited to records where `u_branch == 'EEE'`.

- FR-04 (UI Consistency): ACL rules must function seamlessly across Form Views, List Views, and Service Portal widgets.

4. Non-Functional Requirements

- NFR-01 (Performance): Script execution must be sub-millisecond per record to prevent lag during list view rendering.
- NFR-02 (Security & Compliance): System must default to deny-access if evaluation fails or throws an exception.
- NFR-03 (Maintainability): Scripts must follow clean code standards, avoiding hardcoded sys_ids and expensive database queries.

5. Requirement Traceability Matrix (RTM)

Req ID	Requirement Description	Operation	Target Role	Field Condition
RTM-01	View Branch Records	READ	bb1, bb2, bb3, bb4	Dynamic branch match
RTM-02	Create New Institution Record	CREATE	bb2, bb3, bb4	Valid branch field input
RTM-03	Edit Active Institution Details	WRITE	bb3, bb4	<code>state != 'Closed'</code>
RTM-04	Delete Restricted Branch Record	DELETE	bb4	<code>u_branch == 'EEE'</code>

6. Assumptions, Constraints & Dependencies

- Assumptions: All users are correctly mapped to their respective roles (bb1–bb4) in the platform environment.
- Constraints: No direct database modifications allowed outside the ServiceNow ACL engine framework.
- Dependencies: Schema definition and table structure for `u_institution_details` must be finalized prior to deployment.